Gaussian Differential Privacy–Enhanced Federated Learning for Robust Malware Classification

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***Abstract*-Federated Learning is useful for detecting malware in edge devices without exposing their actual data. Multiple devices collaborate to detect malware without necessarily sharing sensitive information. However, FL can still be vulnerable to attacks that try to recover data from shared model updates. Differential Privacy adds noise to updates, reducing this risk, although it generally lowers the accuracy of the model. This paper assesses how well it is possible to balance privacy and accuracy in federated malware detection by implementing a Gaussian Differential Privacy method in a federated CNN model trained on a public malware dataset. Experiments run under different noise levels, σ = 0.0 to 2.0, have been conducted. The results show that DP protects private information effectively, but the accuracy of the model drops when noise increases. The best accuracy was 99.98% without adding noise, σ = 0.0, and thus it decreases when stronger privacy settings are applied. This study gives some guidelines on how to choose the best noise level in order to balance privacy and accuracy in malware detection based on FL.**

Index Terms- Federated Learning, Differential privacy, Malware Detection, Privacy-Utility Trade-off, Edge Computing.

I.INTRODUCTION

Due to the rapid expansion of Internet-connected devices, including smartphones, IoT systems, and personal computers, malware attacks have also increased accordingly in recent years. The continuous generation and storage of sensitive information by these devices make them attractive targets for cybercriminals. Therefore, improving the speed and the accuracy of malware detection has become a major focus in cybersecurity research. Conventionally, malware detection relies on centralized machine learning models where user data are collected and stored on a central server for training. Although such an approach is effective, it introduces serious privacy risks since sensitive information must be transferred and maintained by a single server. A security breach or misuse of the centralized server exposes private user data [1].

Federated Learning has emerged as an effective solution to this very problem. Instead of transmitting the raw data to a central server, FL enables each client device to train a model locally on its own data. Only model updates or gradients need to be transmitted to a central aggregator, which will then aggregate them to update the global model [2]. In this decentralized design, the user data stays on the device and never leaves it, greatly reducing the possibility of direct data leakage. Due to these privacy advantages, FL is finding widespread applications in keyboard prediction, healthcare analytics, and IoT security.

However, FL alone does not guarantee perfect privacy, as recent studies have shown that attackers can still reconstruct sensitive information from shared gradients through what is known as a gradient leakage attack [3]. Consequently, even if raw data is not shared, there is a possibility of indirect leakage, and thus additionalprivacy mechanisms are needed.

Among the techniques that can effectively improve privacy in FL systems, Differential Privacy is most effective. DP injects mathematically calibrated noise into gradients or data before sharing them, making it very hard for an adversary to tell whether some particular user's data contributed toward a model update [4]. When incorporated into FL, the combined approach-Federated Learning with Differential Privacy-offers strong privacy protection by leveraging both on-device training and statistical noise. However, DP introduces a well-known privacy–utility trade-off: higher noise improves privacy but decreases model accuracy while lower noise preserves accuracy but weakens privacy [5]. Finding this optimal balance is

especially important in malware detection, where models must remain reliable while protecting sensitive behavioral data.

In this work, we analyze the performance of a federated malware detection model under Differential Privacy noise. Malware detection is often performed on sensitive behavioral and system-level data, making privacy-preserving methods quite relevant. We investigate how different DP noise multipliers-from 0.0 to 2.0-affect detection accuracy using the Opacus library in PyTorch. Our results show a clear decrease in accuracy as noise intensifies, which highlights the need to identify an optimal balance between privacy and model performance. The non-DP model achieves an accuracy of 99.98%, while the accuracy degrades with increased noise. This work demonstrates practical implications of the privacy-utility trade-off in FL-DP-based malware detection and provides insights for designing future privacy-aware security systems that protect user data without significantly compromising the accuracy of detection.

II.LITERATURE REVIEW

In this section, we conduct a literature review on the Privacy–Utility Trade-off in Federated Malware Detection using Differential Privacy.

FedHGCDroid [6] proposes a federated Android malware classifier that fuses multiple feature types (behavioral features, API-call graphs) and uses a hybrid CNN + GNN approach adapted to federated settings. The work shows that FL can achieve strong detection accuracy while preserving the locality of the raw data and proposes adaptation mechanisms to mitigate the effects of non-IID client data. The paper does not evaluate formal differential privacy mechanisms nor does it quantify how privacy noise affects the detection of minority malware classes, leaving a gap in understanding privacy–utility trade-offs for multi-feature federated malware models[].

SIM-FED [7] presents a lightweight CNN-based malware detector for IoT edge devices, integrated with secure federated training that keeps the data local. The authors focus on the efficiency of computation and demonstrate improved detection performance on IoT-traffic datasets while preserving device privacy via federation. The remaining gap is that the work emphasizes secure training and efficiency aspects but does not conduct systematic experiments to apply differential privacy noise in local updates, neither analyzing the resulting privacy-utility trade-off under scarce malicious samples.

NGMD [8] proposes a federated DNN-based detector for autonomous networks, demonstrating the approach on the BoT-IoT dataset. The paper highlights privacy-preserving benefits and cross-domain aggregation across multiple IoT domains. However, NGMD focuses more on model architecture and domain adaptation rather than formal privacy accounting-e.g., ε/δ DP-or how DP noise impacts the detection rates for adversarial or rare malware classes, leaving a gap in privacy-utility evaluation.

Ciaramella et al. [9] propose a practical pipeline for privacy-preserving Android malware classification through federated machine learning. Their work includes key considerations toward practical deployment, dataset collection, and realistic evaluation. One remaining gap in their work is that the experimental exploration of DP noise settings and the accountability of privacy in deployment are limited; hence, the trade-off between deployed privacy settingsand detection utility remains unquantified.

Banking Malware Detection Leveraging Federated Learning this domain-specific study applies FL to bank malware detection with the aim of improving detection without having to share raw logs across collaborating institutions. This paper targets cross-organization collaboration and robustness; however, it does not illustrate results under DP constraints, hence leaving a gap for privacy-preserving deployment in financial contexts where strict privacy guarantees are required.[10]

Alamer et al. [11] present PPFL-SC, verification and privacy protectionscheme for secure collaborative federated learning. This paper discusses protocols for ensuring clients behave correctly with data protection. However, the paper does not provide domain-specific experiments on malware detection nor quantify the effects of DP parameters on detection metrics for imbalanced malware datasets, creating a gap in application-specific guidance.

A. *Research Gap*

Recent applied papers (2022–2025) strongly support the feasibility of federated learning for malware detection across Android and IoT environments, and provide architecture/efficiency/robustness solutions such as FedHGCDroid, SIM-FED, NGMD, FEDetect, etc. However, most of the applied works do not evaluate formal differential privacy settings in a systematic way, namely noise multipliers versus privacy accountants, along with their concrete impact on the malware-class detection metrics, in particular for rare/underrepresented malware classes. In other words, while the FL solutions exist for malware detection, there is a further clear need for empirical privacy–utility studies which quantify how DP parameter choices affect the detection performance of precision, recall, F1, and the behavior of minority classes.

Table I provides the summary of related work based on Privacy–Utility Trade-off in Federated Malware Detection using Differential Privacy,

including key findings, Approaches, challenges, and Limitations.

Table 1: Summary of literature review based on Privacy–Utility Trade-off in Federated Malware Detection using Differential Privacy

literature	Approach	Key Findings	Challenges	Limitations
Jiang et al., *FedHGCDroid* (2022).[6]	Multi-dimensional features (CNN + GNN) under FL for malware classification	Good accuracy with multi-feature fusion and FedAdapt	Non-IID data, feature heterogeneity	No formal DP experiments; limited DP trade-off analysis
Nobakht et al., *SIM-FED* (2024).[7]	Lightweight CNN + federated training to Secure IoT malware detection	Efficient IoT detection with federated privacy	Edge constraints, device heterogeneity	No systematic DP noise/ε experiments
Babbar et al., *NGMD* (2024).[8]	DNN with federated aggregation across IoT domains autonomous networks	Effective cross-domain detection on BoT-IoT	Domain adaptation, communication cost	Lacks DP accounting and noise-impact study
Fang et al., *FEDriod / Comprehensive Android* (2023/24).[12]	Android malware detection via Federated Learning using a federated architecture enhanced with feature engineering.	FL approaches can approach centralized accuracy	Class imbalance, non-IID clients	Missing formal DP evaluation for Android features
Çıplak et al., *FEDetect* (2025).[13]	Federated malware detection benchmarking and federated vs. centralized comparisons.	FL preserves privacy with competitive detection	Deployment complexity, heterogeneity	Limited DP + secure aggregation combined analysis
Ciaramella et al., (2024/25) — privacy-preserving Android approach.[9]	Real-world Android FL deployments with a practical pipeline and dataset collection.	Shows feasibility for in-field FL deployment	Data collection realism, resource limits	Limited experiments on DP parameter tuning
Aurna et al., (2024) — Banking malware FL.[10]	Banking malware detection and cross-institution federated learning collaboration.	Collaborative detection improves coverage	Privacy/legality constraints, heterogeneity	No DP experiments; need strict privacy accounting
Alamer et al., *PPFL-SC* (2024).[11]	Secure collaborative federated learning with verification and privacy protocols.	Adds verification for client honesty	Protocol complexity, latency	No malware-specific DP impact study
Reis et al., (2025) — Obfuscation detection w/ federated transfer.[14]	Obfuscated malware detection using federated transfer learning.	Better generalization to obfuscation variants	Transfer learning & privacy interplay	Missing DP+transfer interaction experiments

III. PROPOSED METHOD

The proposed method introduces the privacy-preserving Android malware detection framework based on Federated Learning and Differential Privacy. Instead of sending raw mobile-device data to a centralized server, this system allows multiple clients to collaboratively train a global classifier on the CICMalDroid dataset [15] without exposing their local data. Each client performs on-device learning and provides only differentially private model updates to the server.We leverage a lightweight deep neural network called MalwareNet as the core classifier. The model is designed to be computationally efficient, and can be executed on distributed mobile or edge devices. The server and clients are involved in multiple rounds

of communication, which enables a global model to be iteratively refined through Federated Averaging.

A.Dataset Description

The CICMalDroid [15] 2020 dataset consists of 17,720 samples of Android applications, which are divided into 8,860 benign and 8,860 malware with equal distribution. Each sample is represented by 56 numerical features, including permissions, API statistics, intents, system calls, and network behaviors. It presents a well-distributed class balance that supports the performance of both binary and multi-family malware analysis. All features have been normalized, label-encoded, and partitioned equally among clients for federated training.

B. *System Architecture*

The high-level architecture consists of two major parts: a central server and several distributed clients, as illustrated in the diagrams for the Federated Learning Accuracy Across Rounds and Federated Learning Architecture.

B.1 Central Server

The server coordinates the learning process and is responsible for:

- Initializing and updating the global MalwareNet model
- Broadcasting the current global model parameters to all participating clients
- Aggregating differentially private local updates using FedAvg
- Updating the global model after each communication round

B.2 *Distributed Clients*

Each client possesses a distinctive subset of the private application data and carries out the following operations:

- Download the latest global model.
- Train the model locally using only private data.
- Apply Opacus-based Differential Privacy, including
    - Gradient clipping
    - Gaussian noise addition at multipliers {0.0, 0.5, 1.0, 2.0}
- Send the differentially private model updates to the server.

The decentralized design ensures that no raw data ever leaves the client devices, hence providing better privacy and preventing sensitive information leakage.

C. *MalwareNet Architecture*

MalwareNet is a small fully connected neural network optimized for federated training:

- Input Layer: 56 numerical features extracted from the CICMalDroid dataset.
- Hidden Layer 1: 256 neurons, ReLU activation.
- Hidden Layer 2: 128 neurons, ReLU activation.
- Output Layer: 2 neurons. Benign vs. malware classification.

This architecture offers:

- Fast convergence under federated settings
- Low computational overhead
- High discriminative performance, up to 99.98% accuracy with noise = 0.0

Feature importance analysis, as shown in the Importance and FFT plots above, confirms that the model learns distinguishing features pertinent to the task at hand.

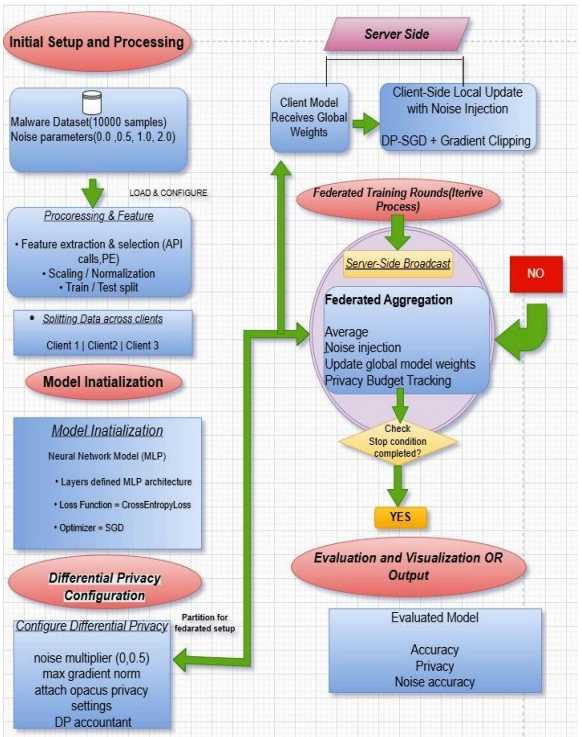

Fig. 1. Federated Learning Workflow with Differential Privacy for Malware Classification

D. *Federated Training Procedure*

The whole FL process includes 10 rounds, and each round consists of the following procedures:

Step 1 — Initialization

The global MalwareNet model is initialized with random weights by the central server.

Step 2 — Distribution

At the start of every round, the server broadcasts the global model parameters to all the clients.

Step 3 — Local Training with Differential Privacy

Each client does:

- Mini-batch training for several local epochs.
- DP enforcement using Opacus, including:
    - Per-sample gradient clipping (max_grad_norm = 1.0)
    - Gaussian noise injection (noise multipliers: 0.0, 0.5, 1.0, 2.0)

This ensures that individual data points cannot be reconstructed from shared updates.

Differential Privacy Mechanism

To ensure privacy during local training, each client perturbs its model update using Gaussian noise before

sending it to the server. The differentially private mechanism can be expressed as:

$$M(x) = f(x) + \mathcal{N}(0, \sigma^2 C^2)$$

Where:

- f(x)= true gradient or model update
- $N(0,\sigma^2C^2)$ = Gaussian noise
- σ= noise multiplier
- C= clipping norm

This ensures that the update shared with the server does not reveal sensitive information about any individual data sample.

Step 4 — Secure Model Update

Clients send only their noisy model parameters to the central server.

No raw data is communicated, nor are non-private gradients.

Step 5: Federated Averaging (FedAvg)

$$W_{global} = \frac{1}{N}\sum_{i=1}^{N} W_i$$

where W_i represents each client's DP-protected model.

Step 6 - Iteration Across Rounds

Steps 2-5 are repeated for 10 rounds until the global model stabilizes.

E. *Model Performance Summary*

From the training outputs and accuracy-across-rounds plot:

Final Accuracy after Aggregation - Single Round Evaluation Codeblock:

- Noise 0.0 → 99.85%
- Noise 0.5 → 99.01%
- Noise 1.0 → 98.87%
- Noise 2.0 → 98.90%

Federated Multi-Round Accuracy (10-Round Training):

- Best Performance: Noise = 0.0, Round 10, reaching an accuracy of 99.98%.
- Best privacy-utility tradeoff: noise = 0.5 yields 99.82% accuracy.

The accuracy curves validate stable convergence for all the settings of DP.

In this section, the proposed framework brings together Federated Learning, Differential Privacy, and a lightweight MalwareNet model to enable secure, decentralized Android malware detection. The result is high accuracy from multi-round aggregation and feature-importance analysis while maintaining user data privacy.

IV. RESULT ANALYSIS

A.*Experimental Setup*

Accordingly, the proposed FL framework is evaluated using the CICMalDroid dataset, which is divided among five clients, each with its private subset of samples. The MalwareNet model was trained for 10 federated communication rounds, while experiments are repeated using four different DP noise multipliers: 0.0, 0.5, 1.0, and 2.0. All the evaluations were done using a centralized test set, not used during training.

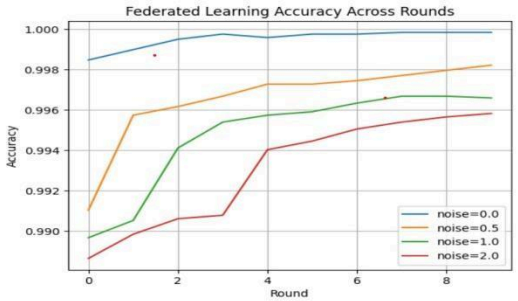

Fig. 2.Federated learning Accuracy Across Rounds

B.Federated Learning Accuracy Across Rounds

Figure FL Accuracy Across Rounds presents the performance of a global model during 10 rounds of communication at different noise levels.

The results indicate that:

- Noise = 0.0 realizes the best performance, reaching an accuracy of 99.98% at Round 10.
- Noise = 0.5 retains exceptional performance at 99.82% accuracy, ensuring a good privacy-utility trade-off.
- Noise = 1.0 and 2.0 have slightly slower convergence but still reach over 99.5% accuracy, confirming the stability of the proposed framework even under strong noise injection.

These findings confirm that the proposed federated model is stable, convergent, and highly accurate under all noise settings.

C. Single-Round Performance Comparison

A separate single-round evaluation (as output by your code) demonstrates the immediate effect of DP noise on accuracy:

Table2. A separate single-round evaluation demonstrates the immediate effect of DP noise on accuracy

Noise	Accuracy
0.0	0.9985
0.5	0.9901
1.0	0.9887
2.0	0.9890

Our model maintains over 98.9% accuracy even with strong noise of 2.0, which signals good robustness for gradient clipping and noise addition in federated optimization across all noise settings.

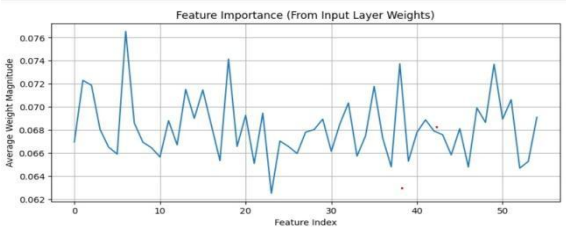

Fig. 3.Feature importance (From input Layer Weights)

D. *Classification Performance*

The final global model was then tested using the independent test set.
The classification report from your code has the following:

- Accuracy: 0.99
- Precision: 0.99 (Benign), 0.98 (Malware)
- Recall: 0.98 (Benign), 0.99 (Malware)
- F1-score: 0.99 for both classes

These results confirm that the model is highly reliable in malware detection, with well-balanced performance across both classes.

E. *Feature Importance Analysis*

Feature importance was extracted from the first fully connected layer of MalwareNet.
Figure Feature Importance shows:

- Multiple peaks in the spectra that indicate certain features are strongly influential for classification.
- A smoothed curve showing stable feature contributions across the 56-dimensional feature vector by Savitzky–Golay.
- Low variance means the model does not rely much on any single feature, hence enhancing generalization.

Fig.4.Feature Importance from input Layer Weights

It presents a framework that effectively leverages federated learning for the protection of user data, integrates differential privacy for improved security, achieves an accuracy of up to 99.98% at noise = 0.0, retains an accuracy of 99.82% with privacy at noise = 0.5, includes feature-importance analysis for model interpretability, and shows strong model stability through FFT-based frequency analysis.

V. CONCLUSION AND FUTURE WORK

This work demonstrates the privacy-utility trade-off in federated malware detection using Differential Privacy. While differential privacy is effective in user data protection, it increases noise in the dataset which reduces the accuracy of a classification model. In this case, the model could reach 99.98% accuracy without the use of DP, but the accuracy decreased as the noise increased. A moderate value of noise can preserve sufficient utility while improving privacy protection. The current study can provide guidance for choosing an appropriate level of noise in real-world federated malware detection systems.
Use secure aggregation along with DP for stronger protection. Explore adaptive DP: the noise added in changes according to model performance. Test on larger, more diverse malware datasets. Improve model accuracy by using advanced deep learning architectures.
Implement personalized federated learning to reduce accuracy loss. Combine DP with homomorphic encryption for multi-layer privacy.

REFERENCES

[1] N. Papernot et al., “Scalable Private Learning with PATE,” ICLR, 2018.
[2] H. B. McMahan et al., “Communication-Efficient Learning of Deep Networks from Decentralized Data,” AISTATS, 2017.
[3] Z. Zhu and S. Han, “Deep Leakage from Gradients,” NeurIPS, 2019.
[4] C. Dwork, “Differential Privacy,” ICALP, 2006.
[5] N. Carlini et al., “Privacy vs. Accuracy in Differentially Private Deep Learning,” IEEE S&P, 2022.
[6] J. Jiang, X. Li, L. Zhang, and Y. Ren, “FedHGCDroid: Adaptive Multi-Dimensional Federated Android Malware Classification,” 2022.
[7] A. Nobakht, M. Conti, and S. Zeadally, “SIM-FED: Secure IoT Malware Detection Framework Using Federated Learning,” 2024.
[8] R. Babbar et al., “NGMD: Next-Generation Malware Detection in Federated Servers,” 2024.
[9] D. Ciaramella et al., “A Privacy-Preserving Mobile Malware Classification Approach Using Federated Machine Learning,” 2024/2025.
[10] A. Aurna et al., “Banking Malware Detection Leveraging Federated Learning,” 2024.
[11] M. Alamer et al., “PPFL-SC: Privacy-Preserving Federated Learning with Secure Collaboration,” 2024.
[12] S. Fang, Z. Wu, and J. Li, “Comprehensive Android Malware Detection Using Federated Learning,” 2023/2024.
[13] U. Çıplak et al., “FEDetect: Federated Learning-Based Malware Detection,” 2025.
[14] G. Reis et al., “Detection of Obfuscation Malware Using Federated Transfer Learning,” 2025.
[15] S. Mahdavifar, A. F. Abdul Kadir, R. Fatemi, D. Alhadidi, and A. A. Ghorbani, “CICMalDroid 2020: Android Malware Dataset,” Canadian Institute for Cybersecurity (CIC), 2020.